Functional Requirements - NTS Design

The purpose of this business requirements document is to provide a draft of the creation of a web-based system for the owner, New Driver System (NTS). The system is intended to assist individuals in passing their driving tests at the DMV. The company consists of the owner, the IT officer, a secretary, and several driving instructors.

System Components and Design

Purpose

- The owner of NTS is the client.
- The client has identified a problem where people fail their driving tests at the DMV.
- The client wants to offer a service that prepares customers for their driving test using two methods:
  - Method 1: Online classes and practice tests.
  - Method 2: On-the-road training.
- The client wants to be able to access and print data and reports online from a computer or mobile device.

System Background

- The client wants the system to allow access from anywhere, both online and offline.
  - The system needs to have a create user function that is permitted on either a computer or a mobile device.
  - The system needs to store user information, including login, name, location, and progress.
- Data should be accessible from any computer or mobile device.
  - The user should be able to access the online classes and tests while online using a PC/Mac/Tablet.
  - The user should be able to access the classes and tests using a mobile device.
  - There should be a link on either platform to schedule on-the-road training.
- The client needs to be able to view reports and data online as well as download and print them.
- The system should include role-based access.
  - The system would have user permissions based on roles: instructors, students, the owner, IT officer, secretary, and possible IT maintenance access.
- All changes to data in the system should be audited and tracked, including activity reports.
  - Reports should be run on a schedule and have the ability to run ad hoc.

- The system should be cloud-based, ensuring a higher level of security and ease of maintenance.
- The system should receive updates from the DMV, so tests are kept up to date.

Objectives and Goals

- Customers need to be able to track their progress on the courses and tests.
  - This could be achieved with an online dashboard that should show course progress, test prep, driving lesson progress, etc.
  - Customers should be able to click on/drill down on any completed course material/progress to view content such as missed questions, covered topics, etc.
    - Links to course content should allow for direct to that section of the course material.
- Customers should be able to create a goal timeline based on their next scheduled exam if needed.
- Customers need to be able to reserve driving lessons online, over the phone, or in person, which will be updated on the account.
  - The outcome of their driving lessons should be tracked by instructor feedback and/or grading. The feedback link should be accessible on the dashboard or some other navigation menu.
  - Customers need to be able to cancel or modify appointments online, also available through a navigation menu or dashboard button.
- In-person lessons need to be able to be scheduled by customers, instructors, or the company secretary.
- Customers can select one of three packages.
  - The system should list information on three packages through the enrollment/new user process.
  - Customers should be able to upgrade packages online at any time.
  - Customers should be able to discontinue a package.
  - The system will also need a financial section for package management and payment management.
- The system should manage driver scheduling.
  - Through the user-role process, drivers should be able to access their schedules, input availability, and provide feedback to students.
  - The system should also record who made the appointment, who canceled an appointment, and who modified the appointment last.
  - Reporting functionality for the client is required.
  - The system should manage instructor/customer/assigned car data for each appointment.

- Customers should be able to manage passwords.
    - Customers should be able to reset their passwords automatically, have “forgot password” functionality, and be prompted for new passwords for security reasons every 45 days.
    - Customers need access to manage their profile, such as password updates, address changes, etc.
- The client needs to be able to run scheduled reports, as well as ad hoc reports.
    - The client should also have a dashboard when logging in. Some dashboard items may include the number of active customers, active drivers, and average client success rates, as well as the option to pull more granular reporting on user time-on-task, most viewed course content, etc.
    - The client’s support team needs to be able to pull reports on changes to the database/activity.
- The system needs to have a “Contact Us” function.

Requirements

Nonfunctional Requirements.

- The client will need a cloud-based storage and backup provider.
- The client will need staff to manage updates to the product, such as platform and DMV updates.

Rationale.

The client indicated they would like the system to be cloud-based, but they did not indicate they already had a cloud-based storage and backup provider. They also indicated they wanted to run the business with minimal technical problems. They will either need staff who can manage the updates regarding the platforms and the DMV, or potentially their cloud-based provider has update packages available.

Performance Requirements

- The system needs to run in a web-based environment for PC and Mac.
- The system needs to run as a mobile web application for Android OS and iOS.
- The system should load the web-based page in less than one second on both a computer and a mobile device (Salz, 22 August 2019).
- Both the web-based product for both computers and mobile devices should be updated no less than once per month, even if the updates are only bug fixes (Shatz, 30 June 2020).
- Customers should be able to see real-time updates to their progress with each completion or save.

Rationale.

The client indicates they would like the system to run web-based on either a computer or a mobile device. Up-time, stability, and speed are important factors for web-based applications.

Platform Constraints

- The system should be able to run on Windows, Mac, Android, and iOS, other platforms can come with future releases.
- The system will need a database to catalog the customer profiles, progress, appointments, results, etc.
- Cloud-based storage and backup to ensure uptime and security.

Rationale.

Client indicated he wants the system to run web-based for both computers and mobile devices. The system will need to keep information on customers, instructors, coursework, tests, vehicles, payment information, schedules, etc. Therefore, a database is needed, and backups are required. The IT officer indicated he wants the system to be cloud-based to minimize security risks and downtime. Cloud-based storage and backup achieve the least downtime by minimizing security risks and server loss.

Accuracy and Precision

- The system should distinguish between user roles through login credentialing. Permissions for access to records and change management are determined by user roles.
- Login credentialing is case-sensitive, with password reset requirements every 45 days.
- The IT officer and the owner are notified of any service degradation, outage, or suspected security breach.

Rationale

As the system involves access to driving record materials, financial information for subscriptions/access, etc., ensuring accuracy and security is important. The system permissions are role-based; therefore, access must be controlled, and precision of permissions is paramount.

Adaptability

- Users will be able to update their profiles, such as addresses and appointment schedules.
- Platform updates will be managed through scheduled weekly update checks when bug fixes or other updates to the system need to occur.
- IT officer will need to be able to make updates to the system or other IT staff at a future date.

Rationale.

The IT officer needs to be able to implement changes to update the system. The process needs to be controlled and follow the release management protocol. Platform updates are

expected, and having weekly checks allows for testing and scheduling to ensure code compliance. Users should be able to update their information with appropriate permissions.

Security

- The system must require login credentials of a username and a password combination.
  - Customers should have some sort of Two-factor authentication (Descalo, 11 January 2022).
  - Staff login would require Two-factor authentication (Descalo, 11 January 2022).
  - Password resets should be mandated every 45 days and must contain letter/number/special character/length requirements (Descalo, 11 January 2022).
  - Customers should have access to password resets with a forget password function; however, there needs to be security questions and possibly two-factor authentication added to the functionality, not a simple send link to email.
  - Login attempts should be limited to five before the time-out and forget password option (Descalo, 11 January 2022).
  - Prevent any previously used passwords in the system.
  - Username cannot be the email address used in profile creation.
- The system would differentiate role access based on the login used, i.e. customer, secretary, instructor, IT officer, and owner.
- The system would have an SSL certificate, and data exchange would be encrypted.
- Cloud-based storage and backup add a layer of security and protection from downtime (Wasabi Technologies, 2023).
- The system sends a notification to the IT officer and the owner for suspected "brute force" hacking attempts.
  - Ex: Multiple customer lockouts in a specified amount of time, lockouts for personnel.

Rationale.

Security is important given the customer and financial information that will be stored in the system. Permissions to the system must be controlled by role-based access. Two-factor authentication adds another layer of security to the system. Utilizing cloud-based storage and backup will mitigate downtime and add additional security. Notifications to the IT officer and owner allow for awareness and decision-making if further action is required.

Functional Requirements

- The system shall validate user credentials when logging in.
- The system shall allow new users to create a profile, one profile per name/dob combination.
- The system shall provide access to the training course.

- The system shall allow for access to previously completed course material.
- The system shall allow for in-person lesson scheduling by the customer, instructor, or secretary.
- The system shall maintain user profile data, including progress, appointments, and pick-up location as needed.
- The system shall keep track of instructor/car scheduling.
- The system shall allow for report download and/or printing by the owner or IT officer.
- The system shall allow for updates from DMV regarding rules, policies, and sample test questions.
- The system shall have a three-tiered access/subscription plan.
- The system shall process subscription/payment information for use.
- The system shall allow customers to see their training progress.
- The system shall allow for instructor feedback on customer records.
- The system shall have a link/button to direct to a “Contact Us” form, which should also include contact information for the company, including address, phone number, and general email address.
- The system needs to have the ability to have packages discontinued at the owner's discretion.

Rationale.

The system needs to allow for secured login, user profile creation, tiered access, payment processing, access to course materials from the computer or mobile devices, reporting for the owner and IT officer, scheduling of in-person lessons by the customer online, the secretary over the phone, or with the instructors themselves. Given the in-person lessons, the system needs to maintain a record of the instructors and cars used for lessons and cataloged by user role. Customers need to see their progress with their coursework, in-person lessons, and sample tests.

User Interface

- The user interface should be determined by the user role.
  - Customers should see their dashboard, including scheduling, coursework, testing, profile, tier plan access, and progress.
  - Drivers should see their students, schedule, and car assignments.
  - The owner should see the dashboard, including ad hoc reporting, basic up-time, customer numbers, and usage, along with financial information.
  - The IT officer should have a dashboard for reporting, up-time, customer numbers, notifications of updates, and usage.
  - Secretary needs to be able to create customer profiles, manage appointments and scheduling, respond to inquiries, log phone calls, etc.

- All users should be able to access their dashboard through a browser on a computer or mobile device. The functionality should be the same for computers or mobile devices.
- The owner and IT officer should be able to download and print ad hoc and prefabricated reports.

Rationale.

The user interface needs to be role-based. The basic interface would be the same for each role; however, different functionality would be present based on the user level. This interface needs to be able to be viewed both on the computer as well as a mobile device as seamlessly as possible.

Assumptions

- Coordination of notifications and updates from the DMV.
- Users must have internet access.
- Customers must have access to payment by credit card.
- Financial data links to a business account for NTS.
- Assumes the customer approves the design at the meeting on March 11th without additional changes.

Rationale.

The client indicated they wanted to receive updates from the DMV. This assumes there is an agreement between NTS and the DMV for those notifications and updates. Internet access is required for system access. The client indicated they wanted to take payment by credit card but did not mention PayPal/Venmo/Zelle, etc.

Limitations

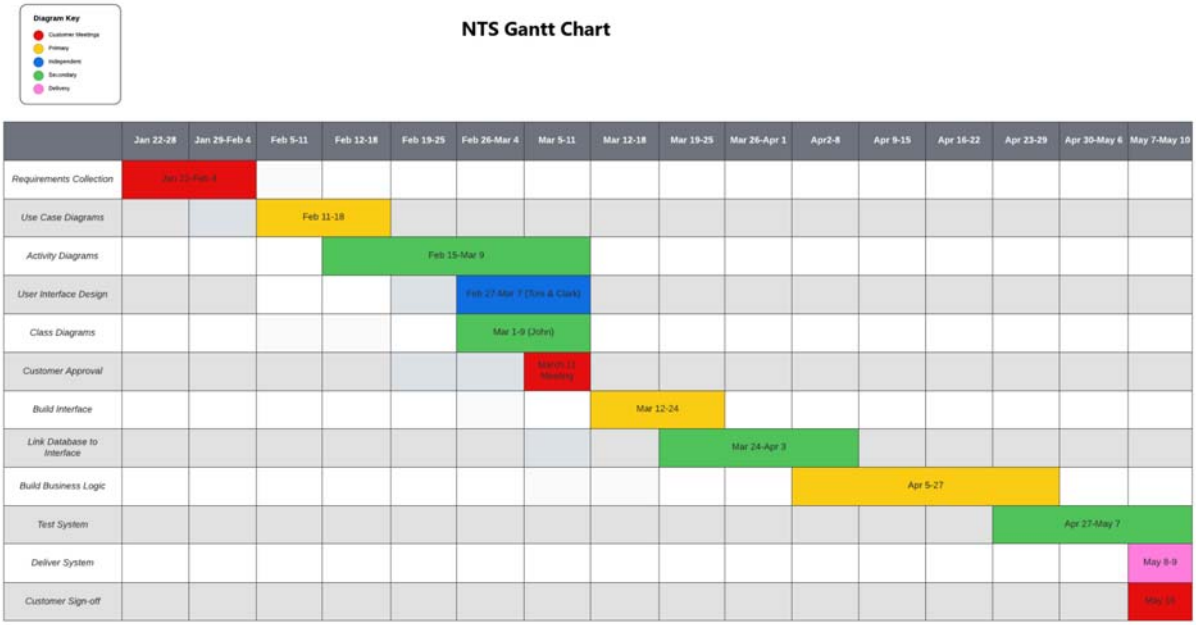
- Updates to code based on DMV changes.
- Potential resource limitation with staff out of the office for six weeks.
- The system may not be accessible during internet outages.

Rationale.

The timeline below is built knowing staff will be out of the office for six weeks; however, it does not have any other staff outages covered. The client indicated they want to be able to update the system based on changes at the DMV. Those changes are not automatically updated; they rely on notifications from the DMV, and someone may have to code changes to the system. The system will be cloud-based, but users will need internet capabilities to access the system.

Gantt Chart

Draft project timeline below.



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